

University of Cape Town
University Examinations – May/June 2023
Department of Mathematics and Applied Mathematics
MAM3000W

SOLUTIONS Module 3MS: Metric Spaces

Time : 2 hours

Full Marks: 40

1. (a) For $x, y \in \mathbb{R}$:
(M1) $d(x, x) = 0$ by definition. If $d(x, y) = 0$, then $x = y$ (since the condition $|x| + |y| = 0$ would give $x = y = 0$).
(M2) For $x \neq y$, $|x| + |y| = |y| + |x|$ so $d(x, y) = d(y, x)$. For $x = y$, $d(x, y) = 0 = d(y, x)$.
(M3) We check that $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in \mathbb{R}$.
If $d(x, z) = 0$, nothing needs to be checked, since d maps into the non-negative reals. If x, y, z are all different, then $d(x, z) = |x| + |z| \leq |x| + |y| + |y| + |z| = d(x, y) + d(y, z)$. If $y = x$ but $y \neq z$, then $d(x, z) \leq d(x, y) + d(y, z)$; in fact, $d(x, z) = d(x, y) + d(y, z)$ since $d(x, y) = 0$. The case where $y \neq x$ but $y = z$ is similar.
(b) $B(2023; 2030) = \{x \in \mathbb{R} \mid d(x, 2023) < 2030\} = (-7, 7) \cup \{2023\}$.
(c) The constant sequence $(x_n) = (7)$ will do. (So should any suitable eventually constant sequence; but no others.)
(d) $\text{int}(\mathbb{R} \setminus \mathbb{Q}) = \mathbb{R} \setminus \mathbb{Q}$.
Proof: For any irrational x , let $r = \frac{1}{2}|x|$, then $B(x; r) = \{x\}$. So each singleton set containing an irrational is open. So the irrationals form an open set (being the union of these open singletons), that is, $\mathbb{R} \setminus \mathbb{Q}$ is open and hence $\text{int}(\mathbb{R} \setminus \mathbb{Q}) = \mathbb{R} \setminus \mathbb{Q}$.

[6+2+2+3=13]
2. (a) Suppose $B(a; r)$ is an open ball in a metric space (X, d) . Here $a \in X$, $r > 0$ and $B(a; r) = \{x \in X \mid d(x, a) < r\}$.
We show that $X \setminus B(a; r) = \{x \in X \mid d(x, a) \geq r\}$ is closed.
Take (x_n) a sequence in $X \setminus B(a; r)$ such that $x_n \rightarrow x$ in (X, d) .
We show that $x \in X \setminus B(a; r)$.
For all $n \in \mathbb{N}$, $r \leq d(a, x_n) \leq d(a, x) + d(x, x_n)$.
Taking limits as $n \rightarrow \infty$ gives $d(a, x) \geq r$, as required.

(b) True.

Suppose on the contrary that $f : X \rightarrow Y$ is a continuous surjective function, and Y is disconnected. Then there exists a continuous function $g : Y \rightarrow \mathbf{2}$ which is not constant. Here $\mathbf{2}$ is the two-element metric space with discrete metric. Notice that $h = g \circ f : X \rightarrow \mathbf{2}$ is continuous. Since g was not constant, there exist $y_1, y_2 \in Y$ with $g(y_1) \neq g(y_2)$. Since f is surjective, there exist $x_1, x_2 \in X$ such that $f(x_1) = y_1$ and $f(x_2) = y_2$, hence $h(x_1) \neq h(x_2)$, making h non-constant. This shows that X is disconnected, a contradiction.

[3+3=6]

3. (a) True.

Suppose (x_n) is a sequence in the metric space such that $x_n \rightarrow x$ and $x_n \in F$ for all $n \in \mathbf{N}$. We show that $x \in F$.

Now, for all $n \in \mathbf{N}$, and for all $i \in I$, $x_n \in F_i$.

Since for all $i \in I$, the set F_i is closed, we have $x \in F_i$.

So $x \in \bigcap_{i \in I} F_i = F$, as required.

(b) False.

In $(\mathbf{R}, d_E)$ let A_1 be the set of rationals and A_2 the set of irrationals. Then $A_1 \cap A_2 = \emptyset$, so $\overline{A_1 \cap A_2} = \emptyset$. But $\overline{A_1} \cap \overline{A_2} = \mathbf{R} \cap \mathbf{R} = \mathbf{R}$.

[3+2=5]

4. (a) The claim is that $T_1 \supseteq T_2 \supseteq \dots \supseteq T_n \supseteq \dots$

Proof: If $x_k \in T_{n+1}$ then $k \geq n+1$, so $k \geq n$, so $x_k \in T_n$.

(b) Let $\epsilon > 0$ be given. Use the fact that (x_n) is Cauchy to choose $N \in \mathbf{N}$ such that $k, \ell \geq N \implies d(x_k, x_\ell) < \frac{\epsilon}{2}$.

Then $n \geq N \implies \text{diam } T_n = \sup\{d(x_k, x_\ell) : k, \ell \geq n\} \leq \frac{\epsilon}{2} < \epsilon$.

(c) Use the fact that $\text{diam } T_n \rightarrow 0$.

(d) Since $x \in \overline{T_N}$, every open ball centred at x hits T_N ; so $B(x, \epsilon/2) \cap T_N \neq \emptyset$.

(e) Suppose $n \geq N$. Then $d(x_n, x) \leq d(x_n, b) + d(b, x) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$.

This uses x_n and b being in T_N , as well as $b \in B(x, \epsilon/2)$.

(f) The result is used to ensure that $\text{diam } T_n$ is a real number, for each n .

[2+2+1+2+2+1=10]

5. (a) We note that the inclusion $f^{-1}(f(A)) \supseteq A$ always holds.
 $f^{-1}(f(\emptyset)) = \emptyset$ since direct and inverse images preserve the empty set.
 $f^{-1}(f(X)) \subseteq X$, by definition; combined with the inclusion above, we get equality.
Finite intersections:
Suppose $f^{-1}(f(A)) = A$ and $f^{-1}(f(B)) = B$.
Then $A \cap B \subseteq f^{-1}(f(A \cap B)) \subseteq f^{-1}(f(A) \cap f(B)) \subseteq f^{-1}(f(A)) \cap f^{-1}(f(B)) = A \cap B$.
So $f^{-1}(f(A \cap B)) = A \cap B$.
Arbitrary unions:
Suppose $f^{-1}(f(A_i)) = A_i$ for all $i \in I$.
Then $f^{-1}(f(\cup A_i)) = f^{-1}(\cup f(A_i)) = \cup f^{-1}(f(A_i)) = \cup A_i$, since direct and inverse images preserve unions.
- (b) Let $x, y \in \mathbf{R}$ be such that $x \neq y$, then for any $G_1, G_2 \in \mathcal{T}$ with $x \in G_1$ and $y \in G_2$, we have G'_1 and G'_2 finite and so $G_1 \cap G_2 \neq \emptyset$ (since $\mathbf{R}$ is infinite and for any $z \notin G'_1 \cup G'_2, z \in G_1 \cap G_2$)

[3+3=6]